

Software Verification and Validation

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Table of Contents

AI in Software Testing: The New Standard

AI in software testing is no longer experimental — it is now a core engineering practice.

- 88% of organizations have adopted AI across business functions (McKinsey 2025)
- 65% of QA teams already use AI in testing workflows (TestRail survey)
- Shift from “whether to adopt” → “how to implement effectively”

Key focus: Faster delivery without sacrificing quality

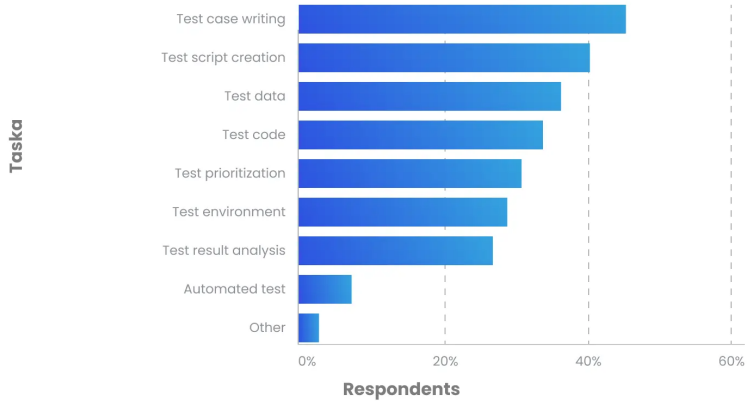
Why Traditional Automation is Not Enough

Traditional automation solved regression testing — but modern systems outgrew it.

- Continuous delivery requires multiple daily releases
- Test suites are too large and expensive to maintain
- UI/API changes break scripts frequently
- 30–40% QA effort spent on maintenance

Result: Automation is no longer sufficient without intelligence

QA tasks most often handled with AI



What AI Brings to Software Testing

AI enhances testing across the full QA lifecycle:

- Test case generation from requirements
- Smart test prioritization (risk-based)
- Self-healing automation
- Predictive defect detection
- Visual regression testing

Shift to Agentic AI Testing

Testing is moving from assistance → autonomy.

- AI-assisted: helps write tests faster
- AI-agentic: AI plans, generates, executes, and analyzes tests
- Human role shifts to strategy and oversight

Trend: 33% of enterprise apps will include agentic AI by 2028
(Gartner)

LLM-Powered Test Generation

Large Language Models transform test creation workflows.

- Generate test cases from Jira/requirements
- Produce Gherkin (Given/When/Then) scenarios
- Create automation scripts (Playwright, Selenium, etc.)
- Identify edge cases and negative scenarios

Impact: 40–60% faster test creation

Self-Healing Test Automation

AI reduces script fragility through adaptive locators.

- Detects broken locators automatically
- Finds alternative UI elements using ML
- Reduces maintenance overhead significantly

Important: Self-healing is a safety net, not a strategy

AI improves testing efficiency using historical data.

- Predicts defect-prone modules
- Prioritizes high-risk test cases
- Reduces unnecessary test execution
- Improves regression confidence under time pressure

AI expands QA beyond functional testing.

- Visual testing (layout, UI consistency)
- Synthetic test data generation
- Real-time monitoring in CI/CD pipelines
- Early anomaly detection (performance/security)

Where AI Still Falls Short

AI is powerful, but not universal.

- Complex stateful systems are hard to model
- Security testing still requires human expertise
- Exploratory testing cannot be fully automated
- Poor data quality reduces AI effectiveness
- Self-healing can hide real regressions

Key Takeaway

AI is reshaping QA, but not replacing it.

- Automation → Intelligence-driven testing
- Scripts → Adaptive systems
- Execution → Decision support

Winning teams: Combine AI speed with human judgment

What This Means for QA Engineers

A key question: **Will AI replace QA engineers?**

Short answer: No — but it will significantly reshape the role.

- AI replaces the **execution layer** (scripts, maintenance, repetitive tests)
- QA engineers move toward higher-level quality responsibilities

Shift: From test execution → quality engineering and decision-making

Evolving Role of QA Engineers in the AI Era

QA work is becoming more strategic and less manual.

- **Test strategy design** — defining what, why, and how to test based on risk
- **AI output evaluation** — validating AI-generated test cases and scripts
- **Exploratory testing** — uncovering usability gaps and unknown edge cases
- **Quality advocacy** — communicating risk to product and leadership

Despite AI advances, key QA responsibilities remain human-driven.

- Understanding real user behavior and context
- Identifying ambiguous or unexpected failure modes
- Making trade-offs between risk, cost, and coverage
- Translating technical issues into business impact

Key insight: AI supports QA — it does not replace judgment

Future Outlook for QA Engineers

The role of QA is evolving, not disappearing.

- From manual testers → quality engineers
- From script writers → AI system supervisors
- From execution-focused → strategy-focused roles

Winning profile: Engineers who understand both QA principles and AI systems

Source Blog: AI in Software Testing – Industry Analysis

- DeviQA insights and industry research
- McKinsey State of AI 2025
- TestRail World Quality Report 2025

Thank You

Thank you for your attention!

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